

Flat Lonely Runners

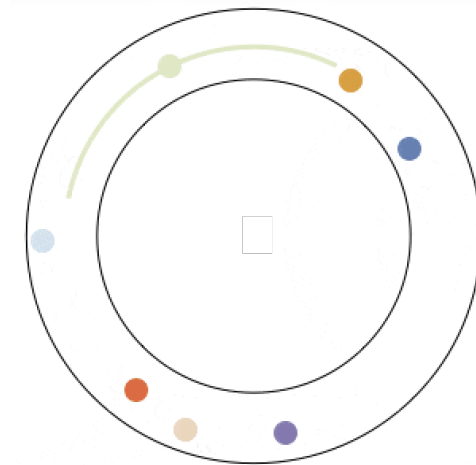
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[arXiv:2609.06259](https://arxiv.org/abs/2609.06259)



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The Lonely Runner Conjecture: From Early Insights to Current Trends and
Future Directions
Rostock 2026

Lonely Runners

Let $\| \cdot \|_{\mathbb{Z}}$ denote distance to the nearest integer and for $\mathbf{n} \in \mathbb{Z}_{>0}^k$ let

$$\text{ML}(\mathbf{n}) := \max_{t \in [0,1]} \min_{1 \leq j \leq k} \|tn_j\|_{\mathbb{Z}}$$

LRC (Wills 1967) $\text{ML}(\mathbf{n}) \geq \frac{1}{k+1}$

Bedert (2026+) $\text{ML}(\mathbf{n}) \geq \frac{1}{2k} + \frac{1}{k^{\frac{3}{2}+o(1)}}$

$\mathbf{n}$ is a **counterexample** if $\text{ML}(\mathbf{n}) < \frac{1}{k+1}$ and a **tight instance** if $\text{ML}(\mathbf{n}) = \frac{1}{k+1}$

Tao (2018) For any k there are only finitely many counterexamples

Improved by **Malikiosis–Santos–Schymura (2025)** and **Giri–Kravitz (2026)**
→ recent records on LRC by **Rosenfeld, Trakulthongchai & Sungkawichai**

Lonely Runner Relations

Let $\| \cdot \|_{\mathbb{Z}}$ denote distance to the nearest integer and for $\mathbf{n} \in \mathbb{Z}_{>0}^k$ let

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Theorem 1 If $\mathbf{n} \in \mathbb{Z}^k$ is a counterexample or tight instance then

- (a) $\mathbf{m} \cdot \mathbf{n} = 0$ for some $\mathbf{m} \in \mathbb{Z}^k$ with $0 < \|\mathbf{m}\|_1 \leq 2k + 3$ and $\|\mathbf{m}\|_1$ odd
- (b) $\mathbf{m} \cdot \mathbf{n} = 0$ for some $\mathbf{m} \in \mathbb{Z}^k$ with $0 < \|\mathbf{m}\|_1 \leq \frac{k+1}{k-1} \text{flt}(k)$

to be defined shortly

Philosophy already present in **Czerwiński (2012)** and **Giri–Kravitz (2026)**

Lonely Fourier Analysis

Theorem 1 If $\mathbf{n} \in \mathbb{Z}^k$ is a counterexample or tight instance then

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Proof Idea Let $h := \frac{k-1}{2(k+1)}$ and define ψ on $\mathbb{T} := [-\frac{1}{2}, \frac{1}{2})$ by

$$\psi(x) := \begin{cases} \sqrt{\frac{2}{h}} \cos\left(\frac{\pi x}{h}\right) & \text{if } |x| \leq \frac{h}{2} \\ 0 & \text{if } \frac{h}{2} < |x| \leq \frac{1}{2} \end{cases}$$

Then

$$W(x) := \int_{\mathbb{T}} \psi(y) \psi(y - x + \frac{1}{2}) dy > 0 \iff \|x - \frac{1}{2}\|_{\mathbb{T}} < h$$

Thus if $\mathbf{n} \in \mathbb{Z}^k$ is a counterexample or tight instance then

$$0 = \prod_{j=1}^k W(n_j t) = \sum_{\mathbf{c} \in \mathbb{Z}^k} (-1)^{c_1 + \dots + c_k} a_{c_1} \dots a_{c_k} e^{2\pi i(\mathbf{c} \cdot \mathbf{n})t}$$

Now analyze $\mathbf{c} \in \mathbb{Z}^k$ with fixed $\mathbf{c} \cdot \mathbf{n}$ according to the parity of $c_1 + \dots + c_k$

Shifted Lonely Runners

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For $\mathbf{n} \in \mathbb{Z}_{>0}^k$ and $\mathbf{s} \in \mathbb{R}^k$ let

$$\text{ML}(\mathbf{n}, \mathbf{s}) := \max_{t \in [0,1]} \min_{1 \leq j \leq k} \|tn_j + s_j\|_{\mathbb{Z}}$$

SLRC (Wills) $\text{ML}(\mathbf{n}, \mathbf{s}) \geq \frac{1}{k+1}$ when $\mathbf{n}$ has distinct entries

Blanco–Criado–Santoro (2026+) counterexamples for $5 \leq k \leq 16$

Poliakova (2026+) counterexamples for $k \geq 95$

Shifted Lonely Runner Relations

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Theorem 2 Let $\mathbf{n} \in \mathbb{Z}_{>0}^k$, $\mathbf{s} \in \mathbb{R}^k$, and $0 < \lambda < \frac{1}{2}$. If $\text{ML}(\mathbf{n}, \mathbf{s}) \leq \lambda$ then $\mathbf{m} \cdot \mathbf{n} = 0$ for some $\mathbf{m} \in \mathbb{Z}^k$ with $0 < \|\mathbf{m}\|_1 \leq \frac{\text{flt}(k)}{1-2\lambda}$.

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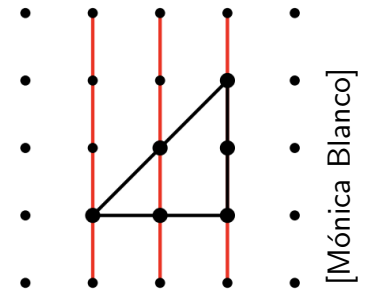
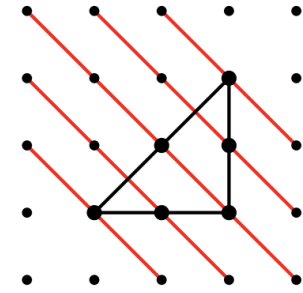
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Corollary Let $\mathbf{s} \in \mathbb{R}^k$ and $0 < \lambda < \frac{1}{2}$. Construct $\mathbf{n} \in \mathbb{Z}^k$ by choosing its k distinct entries from $\{1, 2, \dots, j\}$ uniformly at random. Then the probability that $\text{ML}(\mathbf{n}, \mathbf{s}) \geq \lambda$ tends to 1 as $j \rightarrow \infty$.

Lattice Width & Flatness

The **lattice width** of a convex body $C \subset \mathbb{R}^k$ is

$$\text{width}(C) := \min_{\mathbf{m} \in \mathbb{Z}^k \setminus \mathbf{0}} \max_{\mathbf{a}, \mathbf{b} \in C} \mathbf{m} \cdot (\mathbf{a} - \mathbf{b})$$



[Mónica Blanco]

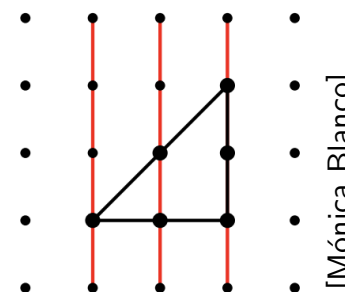
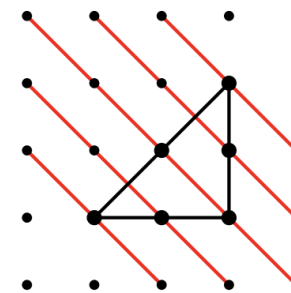
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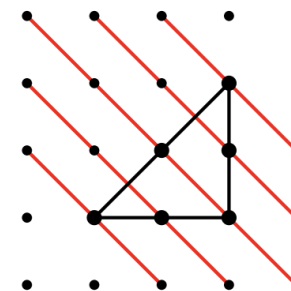
C is **hollow** if C contains no interior lattice points

Khinchin (1948) There exists a (minimal) number $\text{flt}(k)$ such that any hollow convex body $C \subset \mathbb{R}^k$ has $\text{width}(C) \leq \text{flt}(k)$



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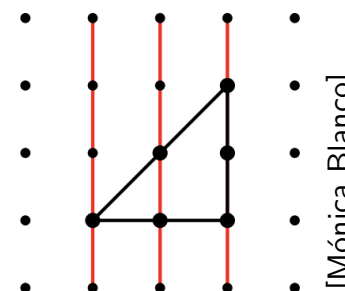
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Reis–Rothvoss (2023) $\text{flt}(k) \in O(k \log(2k)^3)$

Banaszczyk (1996) For centrally symmetric bodies, $\text{flt}(k) \in O(k \log k)$

Codenotti–Santos (2020) There exist k -dimensional polytopes with lattice width larger than k

Lonely Runner Geometry

For $\mathbf{n} \in \mathbb{Z}_{>0}^k$ and $\mathbf{s} \in \mathbb{R}^k$ let

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$$\text{ML}(\mathbf{n}, \mathbf{s}) \geq \lambda \iff t\mathbf{n} + \mathbf{s} = \mathbf{m} + [\lambda, 1 - \lambda]^k \text{ for some } t \in \mathbb{R} \text{ and } \mathbf{m} \in \mathbb{Z}^k$$

$$\iff \mathbf{s} - [\lambda, 1 - \lambda]^k + \mathbb{R}\mathbf{n} \text{ contains an integer point}$$

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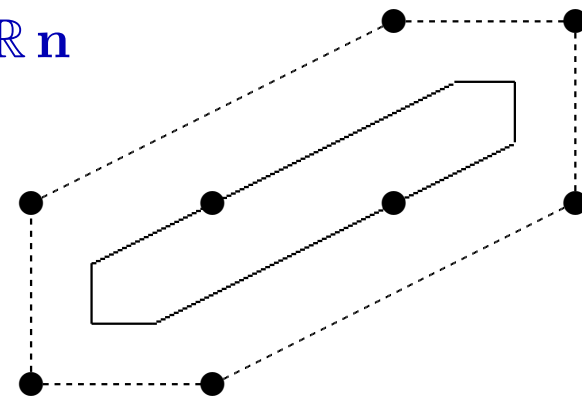
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Henze–Malikiosis (2017) Study the geometry of the zonotopes

$$Z_{\lambda, \gamma}(\mathbf{n}, \mathbf{s}) := \mathbf{s} + [\lambda, 1 - \lambda]^k + [0, \gamma]\mathbf{n}$$

$$Z_{\lambda, \infty}(\mathbf{n}, \mathbf{s}) := \mathbf{s} + [\lambda, 1 - \lambda]^k + \mathbb{R}\mathbf{n}$$

LRC $Z_{\frac{1}{k+1}, \gamma}(\mathbf{n}, \mathbf{0})$ contains an integer point
($\gamma \geq 1$)



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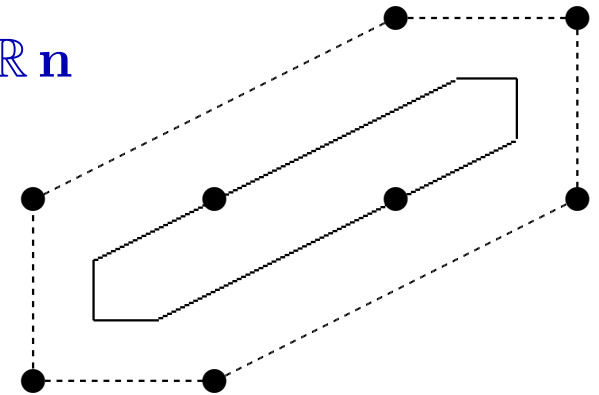
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$Z_{\frac{1}{k+1}, \gamma}(\mathbf{n}, \mathbf{0})$ is hollow ($\gamma \geq 1$) $\iff$

$\mathbf{n}$ is a counterexample or tight instance



Flat Lonely Runners

$$Z_{\lambda,\gamma}(\mathbf{n}, \mathbf{s}) := \mathbf{s} + [\lambda, 1 - \lambda]^k + [0, \gamma] \mathbf{n}$$

$$\begin{aligned} \text{width}(Z_{\lambda,\gamma}(\mathbf{n}, \mathbf{s})) &= \min_{\mathbf{m} \in \mathbb{Z}^k \setminus \mathbf{0}} \max_{\mathbf{a}, \mathbf{b} \in Z_{\lambda,\gamma}(\mathbf{n}, \mathbf{s})} \mathbf{m} \cdot (\mathbf{a} - \mathbf{b}) \\ &= \min_{\mathbf{m} \in \mathbb{Z}^k \setminus \mathbf{0}} (|m_1| \cdots + |m_k|) (1 - 2\lambda) + \gamma |\mathbf{m} \cdot \mathbf{n}| \end{aligned}$$

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If $Z_{\lambda,\gamma}(\mathbf{n}, \mathbf{s})$ is hollow then its lattice width is bounded by $\text{flt}(k)$

→ choose $\gamma = \text{flt}(k)$ and then

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So... if $\text{ML}(\mathbf{n}, \mathbf{s}) \leq \lambda$ then $\mathbf{m} \cdot \mathbf{n} = 0$ for some $\mathbf{m}$ with $0 < \|\mathbf{m}\|_1 \leq \frac{\text{flt}(k)}{1-2\lambda}$

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- ▶ Are there any connections to Schmidt's **subspace theorem** that points of small height in projective space lie in a finite number of hyperplanes? ($\longrightarrow$ back to Diophantine approximation)

